

Vinay Kanakadandila

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience building distributed backend systems and cloud-native platforms at Amazon and Oracle Health. Specialized in scalable microservices, AWS infrastructure, and high-throughput data systems handling 100K+ daily requests. Experienced in distributed systems, large-scale data processing, and AI/ML-driven workflow automation using NLP, LLM-based services, and intelligent decision systems.

TECHNICAL SKILLS

Languages: Java, Python, Kotlin, SQL, JavaScript

Cloud (AWS): Lambda, DynamoDB, EC2, SNS, API Gateway, CloudFormation, CDK, S3, GLUE, Athena, Cloudwatch

Backend & Frameworks: Spring Boot, FastAPI, React, Microservices, REST APIs

Distributed Systems: Caching, Async Processing, Event-Driven Systems, Performance Optimization, API Design

AI/ML: NLP, TF-IDF, Recommendation Systems, LLM Applications, Prompt Engineering

TECHNICAL PROJECTS

PolyBTC-Agent — Autonomous AI Trading Agent | 2026

- Architected an autonomous trading agent combining Claude LLM with quantitative analysis strategies (momentum, mean-reversion) for Polymarket prediction markets.
- Implemented Kelly Criterion risk sizing and circuit breakers to manage downside exposure; deployed on AWS EC2 with Docker for 24/7 operation.

AIJobSearchAgent — Intelligent Job Matching System | 2025

- Built a production Python system that matches job postings using TF-IDF NLP across multiple APIs (Remotive, ArbeitNow, SerpAPI), filtering 1000+ jobs by location and visa requirements.
- Automated end-to-end job aggregation and Excel reporting workflows using Python and SMTP-based notifications.

WORK EXPERIENCE

Amazon, Inc., Austin, TX | Software Development Engineer | 2021 – Present

- Designed and implemented a distributed configuration-tracking platform for 200K+ warehouse network devices, enabling reliable deployment auditing and reducing operational misconfiguration incidents through automated diff analysis tooling.
- Optimized backend microservices by improving database access patterns, introducing caching strategies, and reducing p99 latency for high-volume operational workflows.
- Enhanced service observability through CloudWatch metrics, structured logging, and operational dashboards.
- Mentored junior engineers through code reviews, onboarding, and system design discussions.

Oracle Health (formerly Cerner), Kansas City, MO | Software Engineer | 2015 – 2021

- Spearheaded development of the Registration component for the RevenueCycle product, serving 500+ healthcare organizations and reducing transaction latency by 25% via SQL query optimization and connection pooling.
- Built scalable backend services for a healthcare environment with strict reliability and compliance requirements.

Equity Trust Company, Westlake, OH | Junior Java Developer | 2014 – 2015

EDUCATION

Master of Science, Electrical and Computer Engineering | University of Houston | 2014

Bachelor of Technology, Electrical and Comp | Jawaharlal Nehru Technological University | 2012